honeywell_6000_diagnostic_sop_v1.json

```
{
    "id": "SOP-HW-6000-001",
    "title": "Honeywell 6000 Series: Central Processor Diagnostic Triage",
    "system": "Honeywell 6000 / DPS 8",
    "component": "Central Processor Unit (CPU) / Logic Modules",
    "author": "SymbioStreams Semantic Bridge",
    "source": "Honeywell Maintenance Library (Bitsavers)",
    "description": "Diagnostic procedure for isolating CPU logic faults and instruction execution errors on the Honeywell 6000 series platform.",
    "steps": [
        {
            "step": 1,
            "action": "Console Alarm Indicator Analysis",
            "details": "When a processor fault occurs, examine the system console alarm lights. Identify if the 'System Fault' or 'Execute Error' light is active.",
            "indicator": "Check the status of the Instruction Counter (IC) at the time of the alarm."
        },
        {
            "step": 2,
            "action": "Fault Register Dump",
            "details": "Retrieve the contents of the CPU Fault Register. This 36-bit register contains flags for parity errors, illegal instructions, and boundary violations.",
            "check": "Bit 0 indicates a Memory Parity error. Bit 5 indicates an Illegal Opcode."
        },
        {
            "step": 3,
            "action": "Run Micro-Diagnostic Routine (TET)",
            "details": "Initiate the Test Execution Toolkit (TET) from the maintenance console. This routine cycles through basic ALU logic and register transfers.",
            "logic": "The TET will report a 'Halt' code if a logic gate or MST-80 circuit board is non-responsive."
        },
        {
            "step": 4,
            "action": "Logic Board Isolation",
            "details": "The diagnostic output will provide a module location (e.g., Row G, Slot 12). Power off the processor sub-frame.",
            "remedy": "Replace the failing logic module or check the wire-wrap backplane for signal continuity issues."
        }
    ],
    "diagnostic_flowchart": [
        {
            "condition": "System Fault light ON",
            "next": "Retrieve Fault Register dump."
        },
        {
            "condition": "Memory Parity Bit active",
            "next": "Troubleshoot Memory Control Unit (MCU) and associated RAM storage."
        },
        {
            "condition": "TET diagnostic pass",
            "next": "Logic gates verified. Investigate intermittent cooling or power fluctuations."
        }
    ]
}
```